Code No: 111AE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year Examinations, September - 2023

ENGINEERING CHEMISTRY

(Common to CE, EEE, ME, ECE, CSE, IT, AE, MIE, AGE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	Explain cementation with suitable example.	[2]
b)	Write any four applications of electrochemical series.	[3]
c)	What is sol-gel method? How nano-materials prepared by this method.	[2]
d)	How Bakelite is prepared? Mention any two applications of Bakelite.	[3]
e)	Write any four differences between scales and sludges.	[2]
f)	Give about reverse osmosis. What are its uses?	[3]
g)	What is knocking? How is it prevented?	[2]
h)	Define calorific value? How GCV and LCV are related?	[3]
i)	What are miscelles? Give its uses.	[2]
j)	What is heat treatment? Write about hardening.	[3]

PART-B

(50 Marks)

2.a)	Write the construction and working of calomel electrodes.	
b)	Describe electrochemical corrosion.	[5+5]
	OR	
3.a)	What is Nernst equation? Mention its applications.	
b)	Write about various constituents and their functions of paints.	[5+5]
4.a)	Differentiate chain growth and step-growth polymerization.	
b)	What are FRPs? How are they classified? Give their applications.	
c)	Define and give significance of flash point and fire point of a lubricant.	[4+3+3]
	OR	
5.a)	How plastics are subjected to fabrication by injection moulding?	
b)	Give the mechanism of conduction in poly acetylene.	
c)	How refractories are classified? Give two examples each?	[3+4+3]

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6.a) Calculate the temporary and permanent hardness of water sample containing 7.3 ppm of $Mg(HCO_3)_2$, 16.2 ppm of $Ca(HCO_3)_2$, 9.5 ppm of $MgCl_2$ and 13.6 ppm of $CaSO_4$.

b) Explain Ion-exchange process? How the exhausted resins are regenerated. [5+5]

OR

7.a) What is complexometry? How hardness of water is estimated by EDTA complexometry.

b) What are the specifications of potable water? Write about calgon and phosphate conditioning of water. [5+5]

8. What is the significance of flue gas analysis? Explain Orsat's method for flue gas analysis. [10]

OR

9.a) Explain briefly about proximate analysis of coal and give its significance.

b) Give an account on fixed bed catalytic cracking. [5+5]

10.a) Explain Lead-Silver system with the help of a phase diagram.

b) What are adsorption isotherms? Write about Freundlich's adsorption isotherms. [5+5]

OR

11.a) Derive an equation for phase rule and explain the terms involved in phase rule equation.

b) Give about the optical properties of colloids. [5+5]

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